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Q1. Write a SQL program on DDL Command.

Ans. CREATE Command –

```
mysql> CREATE DATABASE COLLEGE;  
Query OK, 1 row affected (0.19 sec)
```

```
mysql> USE COLLEGE;  
Database changed  
mysql> CREATE TABLE ATTENDANCE(  
-> ID INT PRIMARY KEY,  
-> FNAME VARCHAR (20),  
-> LNAME VARCHAR (20),  
-> BRANCH VARCHAR (20),  
-> AGE INT (20),  
-> MOBILE_NO INT (20)  
-> );  
Query OK, 0 rows affected, 2 warnings (0.91 sec)
```

ALTER Command –

```
mysql> ALTER TABLE ATTENDANCE  
-> ADD COLUMN YEAR INT;  
Query OK, 0 rows affected (1.32 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE ATTENDANCE  
-> MODIFY COLUMN YEAR VARCHAR(20);  
Query OK, 0 rows affected (1.95 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE ATTENDANCE  
-> DROP COLUMN MOBILE_NO;  
Query OK, 0 rows affected (1.09 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE ATTENDANCE  
-> RENAME COLUMN YEAR TO SEMESTER;  
Query OK, 0 rows affected (0.83 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> ALTER TABLE ATTENDANCE  
-> RENAME TO STUDENTS_DETAILS;  
Query OK, 0 rows affected (0.72 sec)
```

TRUNCATE Command –

```
mysql> TRUNCATE TABLE STUDENTS_DETAILS;  
Query OK, 0 rows affected (0.71 sec)
```

DROP Command –

```
mysql> DROP TABLE STUDENTS_DETAILS;  
Query OK, 0 rows affected (0.27 sec)
```

```
mysql> DROP DATABASE COLLEGE;  
Query OK, 0 rows affected (0.22 sec)
```

Q2. Write a SQL program on DML Command.

Ans. INSERT Command –

```
mysql> INSERT INTO STUDENTS_DETAILS
-> VALUES
-> ('1', 'ANKITA', 'SHRIVAS', 'IT', '17', 'FOURTH'),
-> ('2', 'BHAVNA', 'UPADHYAY', 'IT', '19', 'THIRD'),
-> ('3', 'DIVYA', 'GUPTA', 'CS', '17', 'THIRD'),
-> ('4', 'KIRTI', 'SHARMA', 'IT', '18', 'SECOND'),
-> ('5', 'MANYA', 'GUPTA', 'IT', '18', 'FOUTH');
Query OK, 5 rows affected (0.21 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

UPDATE Command –

```
mysql> UPDATE STUDENTS_DETAILS
-> SET BRANCH = 'CS'
-> WHERE FNAME = 'MANYA';
Query OK, 1 row affected (0.48 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

DELETE Command –

```
mysql> DELETE FROM STUDENTS_DETAILS
-> WHERE ID = 5;
Query OK, 1 row affected (0.19 sec)
```

Q3. Write a SQL program on DQL Command.

Ans. SELECT Command –

```
mysql> SELECT * FROM STUDENTS_DETAILS;
```

ID	FNAME	LNAME	BRANCH	AGE	SEMESTER
1	ANKITA	SHRIVAS	IT	17	FOURTH
2	BHAVNA	UPADHYAY	IT	19	THIRD
3	DIVYA	GUPTA	CS	17	THIRD
4	KIRTI	SHARMA	IT	18	SECOND

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT ID, FNAME, BRANCH  
-> FROM STUDENTS_DETAILS;
```

ID	FNAME	BRANCH
1	ANKITA	IT
2	BHAVNA	IT
3	DIVYA	CS
4	KIRTI	IT

```
4 rows in set (0.00 sec)
```

```
mysql> SELECT FNAME  
-> FROM STUDENTS_DETAILS  
-> WHERE SEMESTER = 'THIRD';
```

FNAME
BHAVNA
DIVYA

```
2 rows in set (0.35 sec)
```

Q4. Write a SQL program on DCL Command.

Ans. In DCL Command, first we have to create a user to GRANT and REVOKE privileges to that user.

```
mysql> create user 'vedi'@'localhost' identified by 'password123';
Query OK, 0 rows affected (0.25 sec)
```

GRANT Command –

```
mysql> GRANT SELECT, INSERT ON college.* to 'vedi'@'localhost';
Query OK, 0 rows affected (0.13 sec)

mysql> SHOW GRANTS FOR 'vedi'@'localhost';
+-----+
| Grants for vedi@localhost |
+-----+
| GRANT USAGE ON *.* TO `vedi`@`localhost` |
| GRANT SELECT, INSERT ON `college`.* TO `vedi`@`localhost` |
+-----+
2 rows in set (0.00 sec)
```

REVOKE Command –

```
mysql> REVOKE INSERT ON COLLEGE.* FROM 'vedi'@'localhost';
Query OK, 0 rows affected (0.17 sec)

mysql> SHOW GRANTS FOR 'vedi'@'localhost';
+-----+
| Grants for vedi@localhost |
+-----+
| GRANT USAGE ON *.* TO `vedi`@`localhost` |
| GRANT SELECT ON `college`.* TO `vedi`@`localhost` |
+-----+
2 rows in set (0.00 sec)
```

After performing GRANT and REVOKE Command, we can drop that user.

```
mysql> DROP USER 'vedi'@'localhost';
Query OK, 0 rows affected (0.17 sec)
```

Q5. Write a SQL program on TCL Command.

Ans. For TCL Command, first we have to create a table then fill it as shown below:

```
mysql> select * from second_year;
```

rollno	name	branch	age	fees
1	Divyanshi	IT	17	1000.00
2	Kalpana	CS	18	500.00
3	Shreya	IT	17	850.00
4	vedika	IT	19	1000.00

```
4 rows in set (0.34 sec)
```

For transaction, let these students transferring their fees in each other's account.

START TRANSACTION Command –

```
mysql> START TRANSACTION;  
Query OK, 0 rows affected (0.00 sec)
```

SAVEPOINT Command –

```
mysql> update second_year  
-> set fees = fees - 100  
-> where name = 'vedika';  
Query OK, 1 row affected (0.11 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> savepoint after_vedika;  
Query OK, 0 rows affected (0.00 sec)
```

ROLLBACK Command –

```
mysql> update second_year
-> set fees = fees + 100
-> where name = 'Kalpana';
Query OK, 1 row affected (0.05 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO SAVEPOINT after_vedika;
Query OK, 0 rows affected (0.01 sec)
```

COMMIT Command –

```
mysql> commit;
Query OK, 0 rows affected (0.19 sec)
```

Table after the transactions-

```
mysql> select * from second_year;
+-----+-----+-----+-----+-----+
| rollno | name      | branch | age  | fees    |
+-----+-----+-----+-----+-----+
| 1      | Divyanshi | IT     | 17   | 1000.00 |
| 2      | Kalpana   | CS     | 18   | 500.00  |
| 3      | Shreya    | IT     | 17   | 850.00  |
| 4      | vedika    | IT     | 19   | 900.00  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

Q6. Write a SQL program on relation between primary and foreign key.

Ans. For Foreign key, we have to make relation between two tables and make Primary key of Table1 reference in Table2 as shown-

```
mysql> USE COLLEGE;
Database changed
mysql> CREATE TABLE COLLEGE;
ERROR 4028 (HY000): A table must have at least one visible column.
mysql> CREATE TABLE COLLEGE(
    -> studentID int primary key,
    -> name varchar (90),
    -> age int);
Query OK, 0 rows affected (1.75 sec)
```

```
mysql> desc college;
```

Field	Type	Null	Key	Default	Extra
studentID	int	NO	PRI	NULL	
name	varchar(90)	YES		NULL	
age	int	YES		NULL	

3 rows in set (0.76 sec)

```
mysql> CREATE TABLE second_year(
    -> roll_no int primary key,
    -> subject_no int,
    -> studentID int,
    -> FOREIGN KEY (studentID) REFERENCES COLLEGE (studentID)
    -> );
Query OK, 0 rows affected (0.69 sec)
```

```
mysql> desc second_year;
```

Field	Type	Null	Key	Default	Extra
roll_no	int	NO	PRI	NULL	
subject_no	int	YES		NULL	
studentID	int	YES	MUL	NULL	

3 rows in set (0.00 sec)

Then, insert values in both the tables.

```
mysql> insert into college values
-> (1, 'aditi', 21),
-> (2, 'bhavna', 17),
-> (3, 'meera', 19);
Query OK, 3 rows affected (0.11 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> insert into second_year values
-> (1,101,1),
-> (2,102,2),
-> (3,103,3);
Query OK, 3 rows affected (0.15 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> select * from college;
+-----+-----+-----+
| studentID | name   | age  |
+-----+-----+-----+
|          1 | aditi  | 21   |
|          2 | bhavna | 17   |
|          3 | meera  | 19   |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

```
mysql> select * from second_year;
+-----+-----+-----+
| roll_no | subject_no | studentID |
+-----+-----+-----+
|        1 |          101 |          1 |
|        2 |          102 |          2 |
|        3 |          103 |          3 |
+-----+-----+-----+
3 rows in set (0.00 sec)
```

Now, if we try to insert value in Table2 or delete from Table1 then it will show error of a foreign key constraints fails.

```
mysql> INSERT INTO second_year VALUES
-> (4,104,4);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key
constraint fails ('college`.`second_year`, CONSTRAINT `second_year_ibfk_1` FOREIGN KEY (`studentID`) REFERENCES `college` (`studentID`))

mysql> DELETE FROM COLLEGE
-> WHERE age='17';
ERROR 1451 (23000): Cannot delete or update a parent row: a foreign
key constraint fails ('college`.`second_year`, CONSTRAINT `second_year_ibfk_1` FOREIGN KEY (`studentID`) REFERENCES `college` (`studentID`))
```

Q7. Write a SQL program on Views.

Ans. First we have to create a table then fill it as shown below:

```
mysql> select * from STUDENTS;
```

id	name	branch	age	semester
1	Divyanshi	IT	17	THIRD
2	Kalpana	CS	18	SECOND
3	Shreya	IT	19	THIRD
4	vedika	IT	18	FOURTH
5	Yashu	CS	20	FOURTH
6	Zania	CS	17	THIRD

```
6 rows in set (0.00 sec)
```

Then create a view-

```
mysql> CREATE VIEW ADULT AS
-> SELECT name, age
-> FROM STUDENTS
-> WHERE age >= 18;
Query OK, 0 rows affected (0.23 sec)
```

Now, we can retrieve the data using select command-

```
mysql> select * from ADULT;
```

name	age
Kalpana	18
Shreya	19
vedika	18
Yashu	20

```
4 rows in set (0.13 sec)
```

Q8. Write a SQL program on Normalization.

Ans. For Normalization, first we have to create a main table as shown-

```
mysql> use college;
Database changed
mysql> CREATE TABLE CLASS(
    -> STUDENT VARCHAR (90),
    -> TEACHER VARCHAR (90),
    -> SUBJECT VARCHAR (90)
    -> );
Query OK, 0 rows affected (0.57 sec)

mysql> INSERT INTO CLASS VALUES
    -> ('Shreya', 'Nidhi', 'DBMS'),
    -> ('Shreya', 'Nidhi', 'WT'),
    -> ('Vedika', 'Veena', 'OS'),
    -> ('Vedika', 'Nidhi', 'DBMS');
Query OK, 4 rows affected (0.40 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> select * FROM CLASS;
+-----+-----+-----+
| STUDENT | TEACHER | SUBJECT |
+-----+-----+-----+
| Shreya  | Nidhi   | DBMS    |
| Shreya  | Nidhi   | WT      |
| Vedika  | Veena   | OS      |
| Vedika  | Nidhi   | DBMS    |
+-----+-----+-----+
4 rows in set (0.00 sec)
```

Now, we see that the table is not in normal form as it has repetition in columns.

To make the above table in normal form, we have to removing repetitive values by splitting this table into two tables .

Now, create those two tables-

```
mysql> CREATE TABLE C1(  
-> TEACHER VARCHAR(90),  
-> SUBJECT VARCHAR (90)  
-> );  
Query OK, 0 rows affected (0.55 sec)  
  
mysql> CREATE TABLE C2(  
-> STUDENT VARCHAR (90),  
-> TEACHER VARCHAR (90)  
-> );  
Query OK, 0 rows affected (0.72 sec)
```

Then insert values in it.

```
mysql> INSERT INTO C1 VALUES  
-> ('Nidhi', 'DBMS'),  
-> ('Nidhi', 'WT'),  
-> ('Veena', 'OS');  
Query OK, 3 rows affected (0.41 sec)  
Records: 3 Duplicates: 0 Warnings: 0  
  
mysql> INSERT INTO C2 VALUES  
-> ('Shreya', 'Nidhi'),  
-> ('Vedika', 'Veena'),  
-> ('Vedika', 'Nidhi');  
Query OK, 3 rows affected (0.19 sec)  
Records: 3 Duplicates: 0 Warnings: 0
```

Now, the below two tables are in normal form.

```
mysql> SELECT * FROM C1;  
+-----+-----+  
| TEACHER | SUBJECT |  
+-----+-----+  
| Nidhi   | DBMS    |  
| Nidhi   | WT      |  
| Veena   | OS      |  
+-----+-----+  
3 rows in set (0.00 sec)  
  
mysql> SELECT * FROM C2;  
+-----+-----+  
| STUDENT | TEACHER |  
+-----+-----+  
| Shreya  | Nidhi   |  
| Vedika  | Veena   |  
| Vedika  | Nidhi   |  
+-----+-----+  
3 rows in set (0.00 sec)
```